

Learner Intelligence System architecture and ERD

Learner Intelligence System ERD

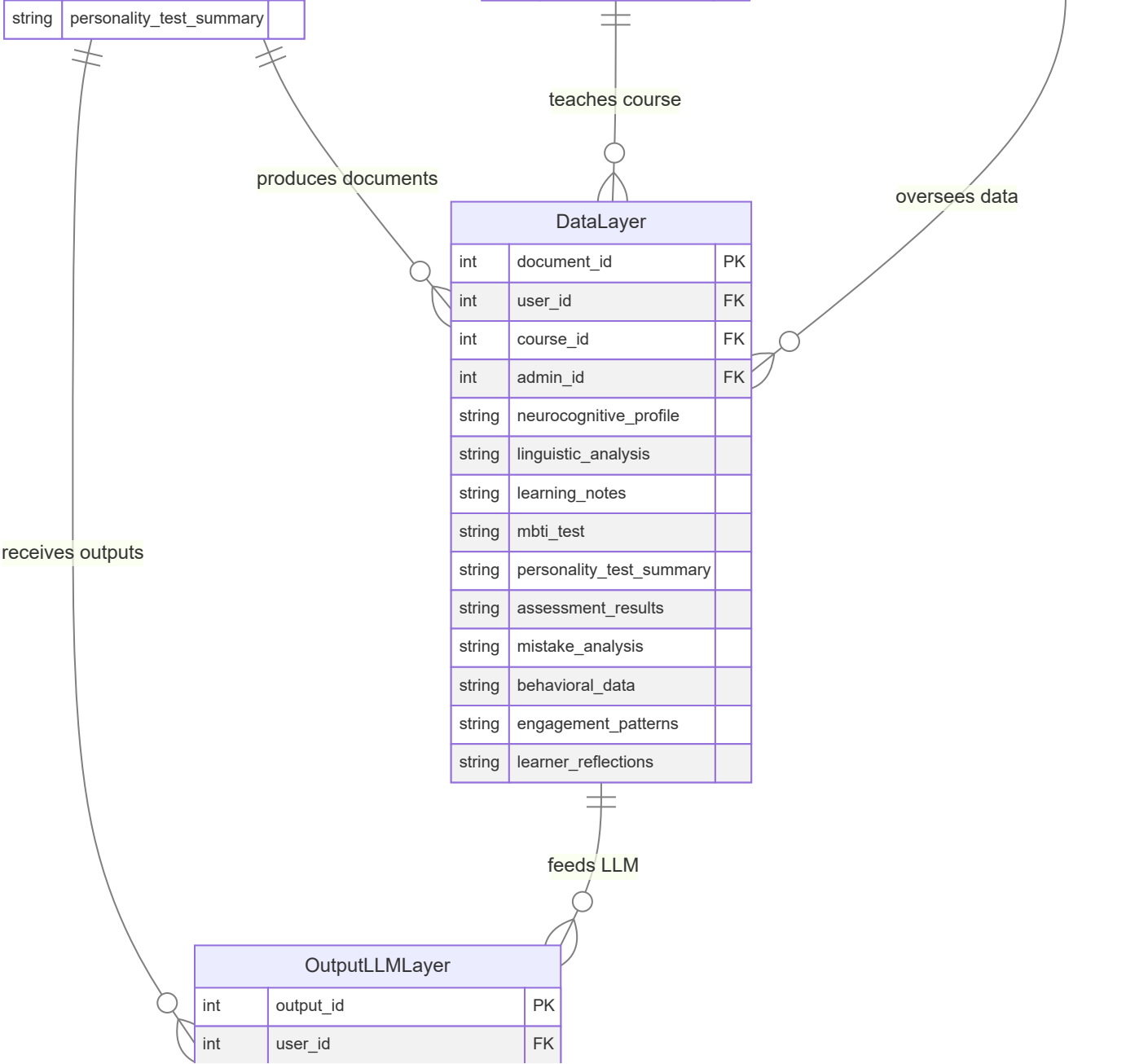
Purpose

This ERD represents the first version of a layered learner-intelligence system. The system collects learner, instructor, and administrative data into a shared data layer. That data layer then feeds an LLM output layer that generates personalized recommendations, learner summaries, course paths, competency guidance, and implementation insights.

UserLayer		
int	user_id	PK
string	name	
int	age	
string	education_level	
string	transition_type	
string	current_status	
string	goals	
string	constraints	
string	interests	
string	cv_reference	
string	transcript_reference	
string	mbti_test_result	
string	personality_test_summary	

InstructorLayer		
int	instructor_id	PK
string	name	
string	role	
int	course_id	
string	course_name	
string	syllabus	
string	modules	
string	assessments	
string	attendance	
string	student_materials	
string	competencies_taught	

AdminLayer	
int	admin_id
string	name
string	role
string	teacher_reviews
string	data_tracking_costs
string	effectiveness_indica
string	outcomes
string	salary_or_career_ot
string	satisfaction_metrics



int	document_id	FK
string	entry_type	
string	title	
string	summary	
string	learner_profile_summary	
string	lesson_or_topic	
string	why_it_matters	
string	top_3_course_paths	
string	recommended_competencies	
string	system_relevance	
string	implementation_implications	
decimal	confidence_score	
string	connections_to_prior_knowledge	
string	user_selected_path	
string	user_feedback	
string	override_reason	

Layer Definitions

User Layer

The `UserLayer` stores learner-level identity, educational background, transition context, goals, constraints, interests, and reference documents. This layer captures the learner's self-reported and externally validated context.

Key examples include:

- Education level
- Transition type
- Current status
- Goals and constraints
- CV and transcript references
- Personality or cognitive-style summaries

Instructor Layer

The `InstructorLayer` stores course-level and instructor-provided context. In this V1 schema, instructor data and course data are grouped together for simplicity.

This layer provides:

- Course structure
- Syllabus information

- Module information
- Assessment data
- Attendance context
- Student materials
- Competencies taught

In a later normalized version, this could be split into separate `Instructor`, `Course`, `Module`, and `Assessment` entities.

Admin Layer

The `AdminLayer` stores administrative and program-level context. This includes institutional effectiveness indicators, satisfaction metrics, teacher reviews, costs, and career outcomes.

This layer is useful for connecting individual learner recommendations to broader program performance and workforce outcomes.

Data Layer

The `DataLayer` acts as the central processing layer. It consolidates learner, course, and administrative inputs into structured records that can be used by the LLM output layer.

This layer includes:

- Neurocognitive profile data
- Linguistic analysis
- Learning notes
- Assessment results
- Mistake analysis
- Behavioral data
- Engagement patterns
- Learner reflections

This is the main bridge between raw input data and AI-generated outputs.

Output LLM Layer

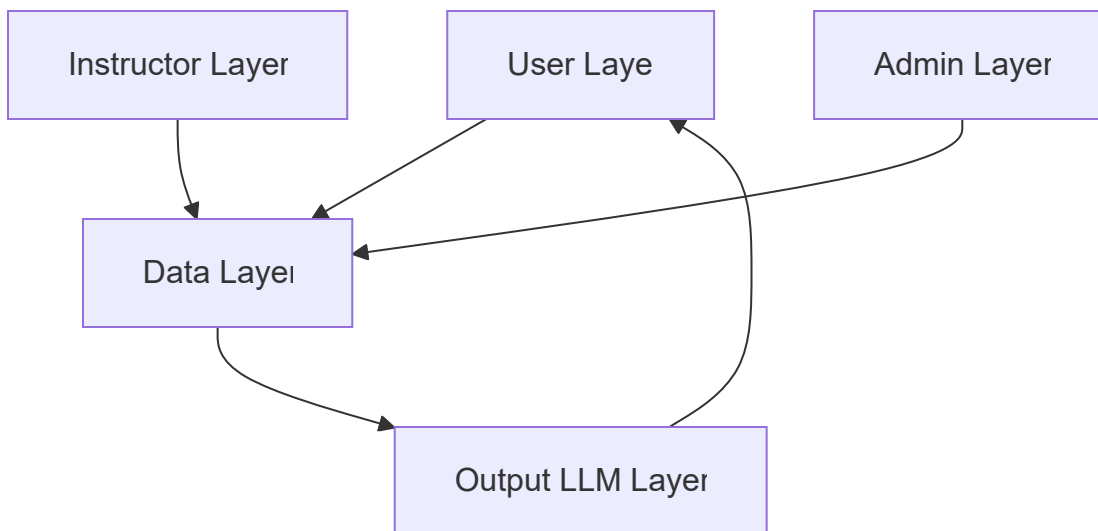
The `OutputLLMLayer` stores the AI-generated outputs created from the data layer. These outputs may include summaries, personalized course recommendations, competency recommendations, relevance explanations, and learner feedback loops.

This layer includes:

- Learner profile summaries
- Lesson or topic summaries
- Recommended course paths
- Recommended competencies
- System relevance notes
- Implementation implications
- User feedback
- Override reasons
- Confidence scores

System Flow

The system follows this high-level flow:



Design Notes

This is a conceptual V1 ERD rather than a fully normalized production schema. The current structure is optimized for understanding the system architecture, not for minimizing redundancy.

The strongest architectural pattern is the separation between:

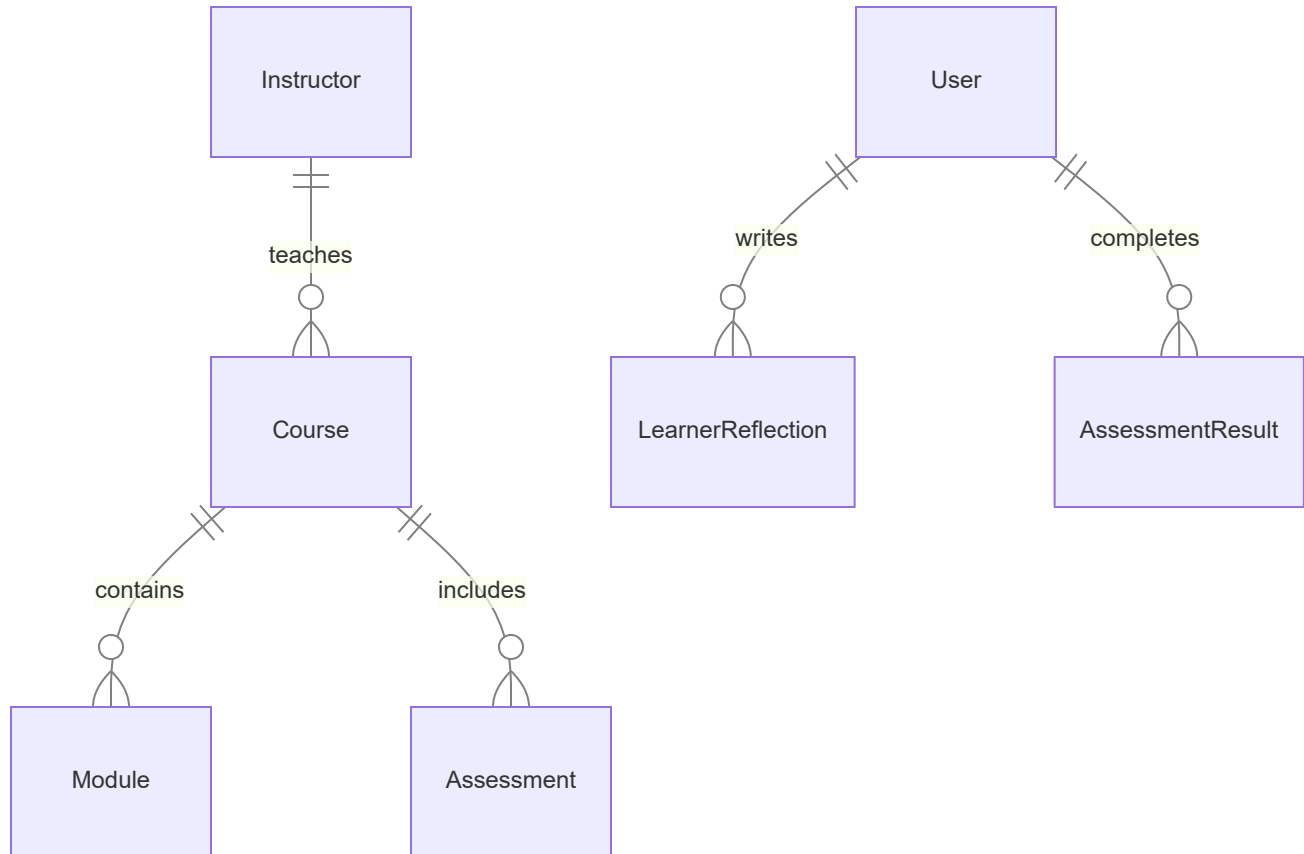
1. **Input layers:** user, instructor, and admin data
2. **Processing layer:** consolidated learner and course data
3. **Output layer:** LLM-generated recommendations and summaries

This structure makes the system explainable because every LLM output can be traced back to underlying learner, course, or administrative data.

Future Normalization Opportunities

In a future production version, several fields could become their own tables.

Potential future entities:



Recommended future splits:

- Split `InstructorLayer` into `Instructor`, `Course`, `Module`, and `Assessment`
- Split `AdminLayer` into `ProgramMetrics`, `OutcomeMetrics`, and `SatisfactionMetrics`
- Split `top_3_course_paths` into a separate recommendation table
- Split `recommended_competencies` into a separate competency recommendation table
- Convert `cv_reference` and `transcript_reference` into foreign keys to a future `DocumentReference` table

V1 Interpretation

This diagram should be interpreted as a layered learner-intelligence architecture.

The core idea is:

Learner, instructor, and administrative data are consolidated into a shared data layer, which then feeds an LLM layer that generates personalized learning guidance, competency recommendations, and system-level insights.